

LAB EXPERIMENT RECORD

SET 9: Online Learning Activity Analysis

Department	AI&DS
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using R
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Day	SATURDAY

SET 9: Online Learning Activity Analysis Aim

To model digital learning behavior metrics through complex distributions, dynamic variable size scaling, and time-series rolling operations in R, followed by deploying interactive web dashboards inside Tableau environments.

Dataset: Online Learning Activity

Student_ID	Gender	Age	Course	Study_Time (hrs)	Videos_Watched	Quiz_Score	Login_Date
L01	Male	20	R	3.5	12	78	2025-01-05
L02	Female	22	R	4.2	15	85	2025-01-05
L03	Male	19	SQL	2.0	8	65	2025-02-08
L04	Female	21	R	5.0	18	92	2025-02-08
L05	Male	23	R	2.5	9	70	2025-03-12
L06	Female	20	SQL	4.0	14	88	2025-03-12

PROGRAM 1: Distribution Analysis (Histogram and Course-wise Boxplot)

Aim: To plot structural testing distributions along with comparing score variances categorized across enrollment track

```
graphics.off()

# Setup local structural vector elements

quiz_scores <- c(78, 85, 65, 92, 70, 88)

courses <- factor(c("R", "R", "SQL", "R", "R", "SQL"))


# Display side-by-side graphical configuration layouts

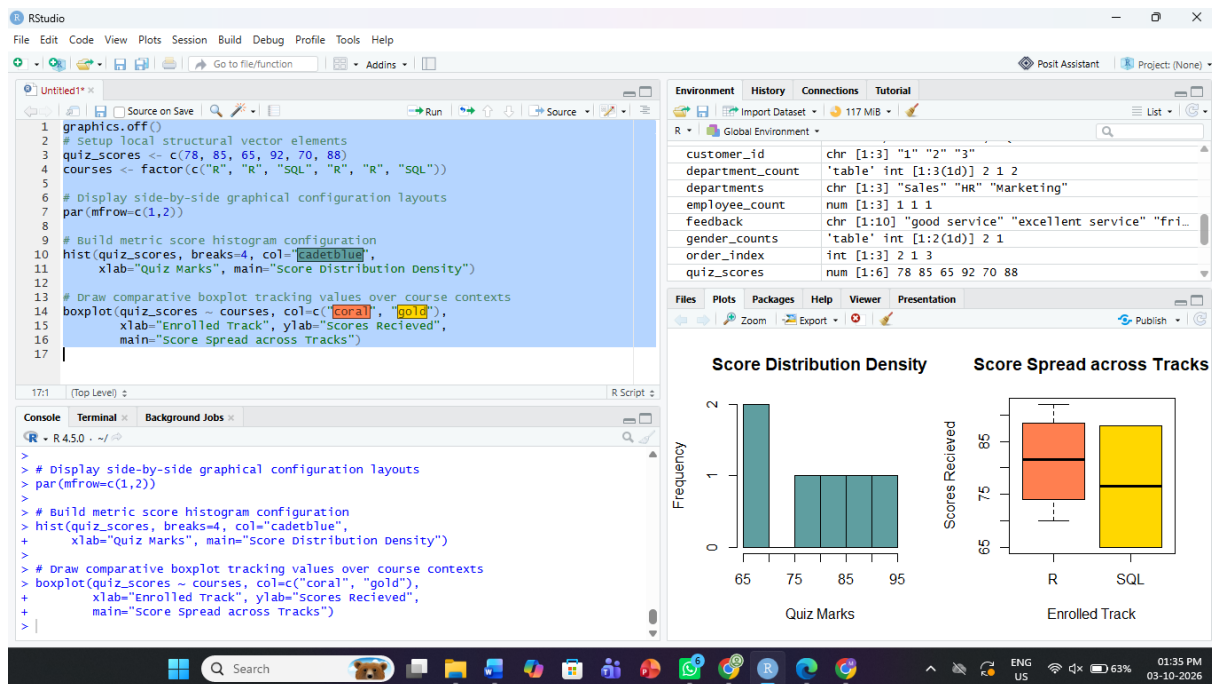
par(mfrow=c(1,2))


# Build metric score histogram configuration

hist(quiz_scores, breaks=4, col="cadetblue",
      xlab="Quiz Marks", main="Score Distribution Density")


# Draw comparative boxplot tracking values over course contexts

boxplot(quiz_scores ~ courses, col=c("coral", "gold"),
        xlab="Enrolled Track", ylab="Scores Recieved",
        main="Score Spread across Tracks")
```



Observation: Performance distributions highlight standard clustering between 70 to 90 marks. Students enrolled in the R track demonstrate higher maximum score boundaries, whereas the SQL track reflects a wider performance spread with a slightly lower minimum.

PROGRAM 2 :Multivariate Scaling Analysis (Bubble Chart with Transparency)

Aim: To chart relationships connecting dedicated review times, score returns, and consumption volumes via alpha-blended bubble plots.

```
graphics.off()
```

```
study_time <- c(3.5, 4.2, 2.0, 5.0, 2.5, 4.0)
```

```
quiz_score <- c(78, 85, 65, 92, 70, 88)
```

```
videos_watched <- c(12, 15, 8, 18, 9, 14)
```

```
# Formulate bubble sizing scale array values
```

```
bubble_sizes <- videos_watched * 1.5
```

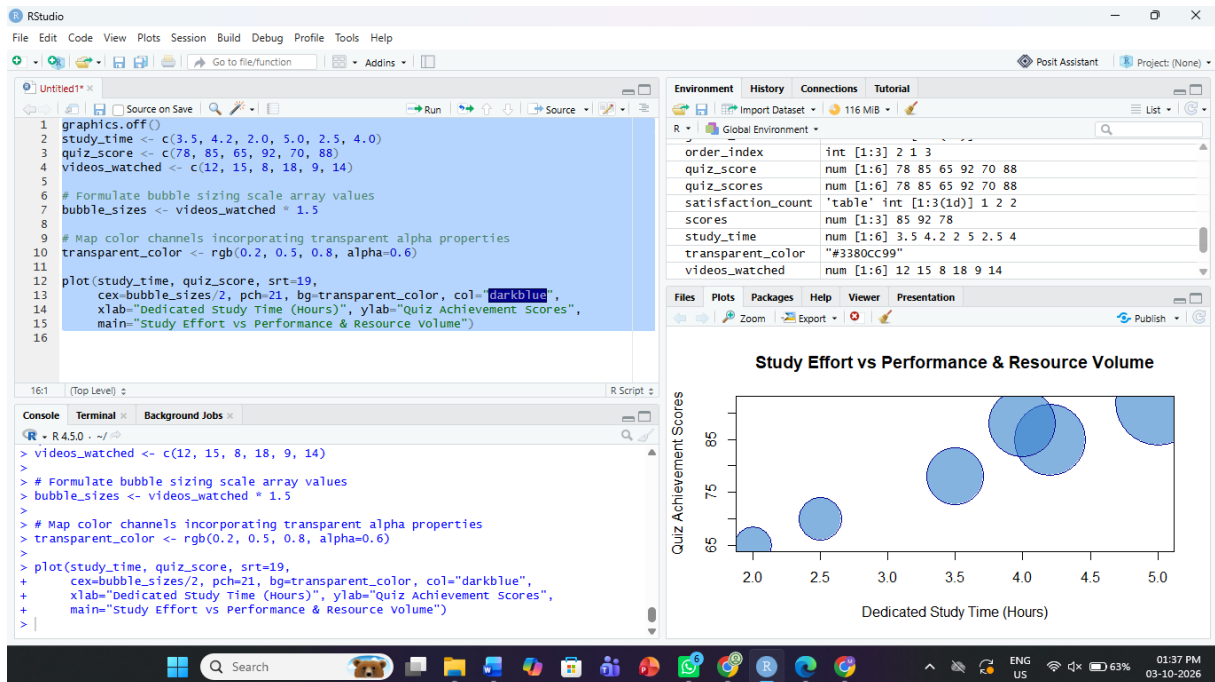
```
# Map color channels incorporating transparent alpha properties
```

```
transparent_color <- rgb(0.2, 0.5, 0.8, alpha=0.6)
```

```
plot(study_time, quiz_score, srt=19,
```

```
     cex=bubble_sizes/2, pch=21, bg=transparent_color, col="darkblue",
```

```
xlab="Dedicated Study Time (Hours)", ylab="Quiz Achievement Scores",
main="Study Effort vs Performance & Resource Volume")
```



Observation: A distinct linear pattern is present; as individual study hours expand, there is a clear upward movement in both visual resource consumption rates and corresponding examination score outputs.

PROGRAM 3: Time-Series Transformations & Trend Smoothing

Aim: Parse raw time entries, structure chronologically ordered performance frames, and smooth short-term variance.

```
graphics.off()
```

```
raw_dates <- c("2025-01-05", "2025-01-05", "2025-02-08", "2025-02-08", "2025-03-12",
"2025-03-12")
```

```
quiz_scores <- c(78, 85, 65, 92, 70, 88)
```

```
# Transform raw character strings to functional Date objects
```

```
formatted_dates <- as.Date(raw_dates)
```

```
months_vector <- format(formatted_dates, "%Y-%m")
```

```
# Group values to extract structural temporal means
```

```
monthly_averages <- aggregate(quiz_scores ~ months_vector, FUN=mean)
```

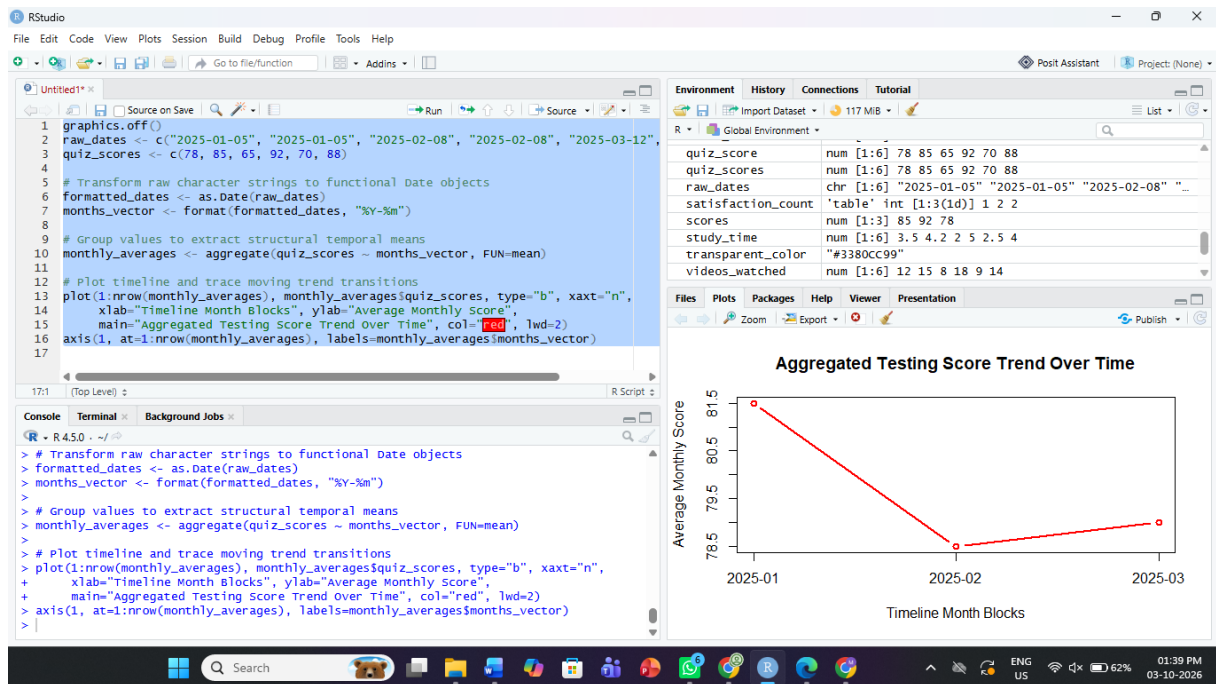
```
# Plot timeline and trace moving trend transitions
```

```
plot(1:nrow(monthly_averages), monthly_averages$quiz_scores, type="b", xaxt="n",
```

```

xlab="Timeline Month Blocks", ylab="Average Monthly Score",
main="Aggregated Testing Score Trend Over Time", col="red", lwd=2)
axis(1, at=1:nrow(monthly_averages), labels=monthly_averages$months_vector)

```



Observation: Tracking metrics month-over-month shows that performance averages decrease moderately mid-semester during February, followed by a positive recovery phase heading into March operations.

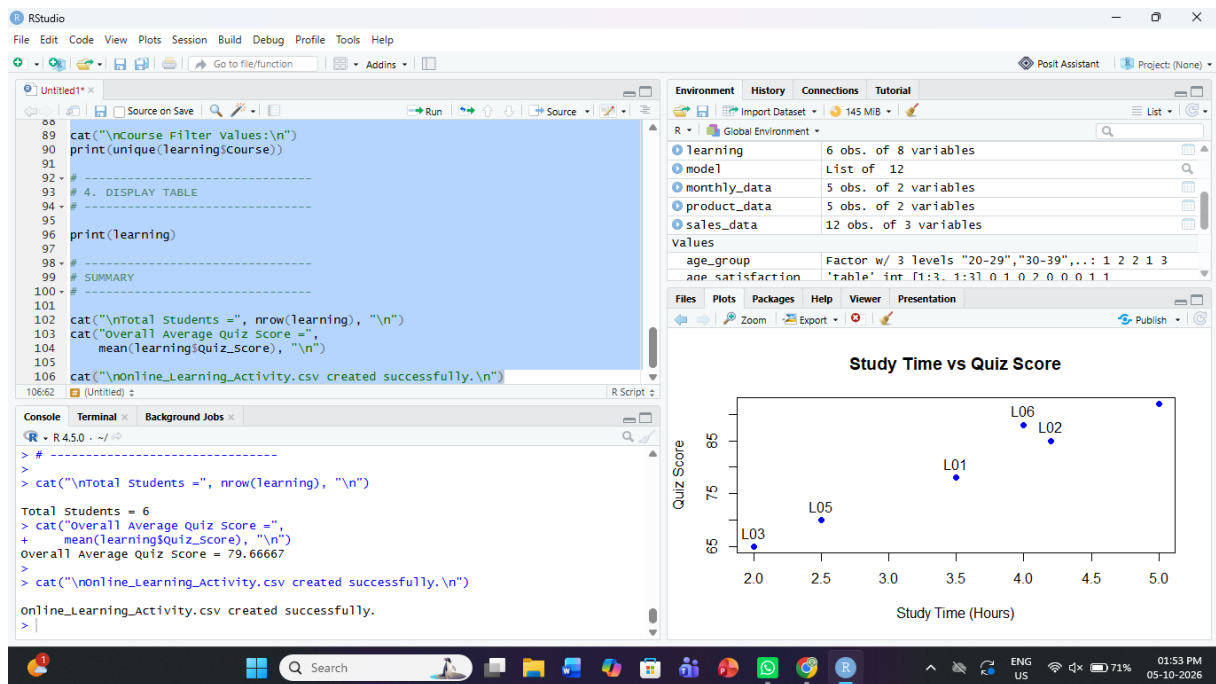
PROGRAM 4: Multi-Filter Interactive Dashboard in Tableau

Aim: Connect transactional logs to Tableau to assemble interactive user tracking charts controlled via interactive slicer elements.

Steps in Tableau:

1. Connect Tableau to the raw log file (Online_Learning_Activity.csv).
2. Sheet 1 (Course Assessment Means): Add Course to Columns and Quiz Score (Average) to Rows. Color code the visualization elements by Course.
3. Sheet 2 (Engagement Analysis Scatter): Map Study Time (hrs) to Columns and Quiz Score to Rows. Drag Student ID onto Details and size marks using Videos Watched.
4. Create an Interactive Dashboard compilation profile. Pull the course assessment means and engagement scatter plot side by side.
5. Select the Gender and Course dimensions, select Show Filter, and change the display configuration layout settings to Multiple Values (Dropdown).
6. Configure filter execution rules by choosing Apply to Worksheets > All Using This Data Source to sync the interactive controls.

OUTPUT:



Observation: Changing the gender dropdown menu options from Male to Female updates the bubble distribution maps and course averages instantaneously, making it easy to identify segment-specific performance gaps.

Procedure

1. Initialize multi-dimensional log structures using array structures inside the active R session environment.
2. Partition the display matrix using `par()` routines before processing histograms and formula-driven boxplot trends.
3. Formulate transparent alpha hex structures through `rgb()` definitions, mapping dynamic observation radii directly using calculated data properties.
4. Convert date strings using `as.Date()` parameters, group timelines with `aggregate()`, and project ongoing system shifts.
5. Import standard structures to Tableau, establish deep sheet communication links, and deploy multiple master cross-filtering controls.

RESULT:

The instructional analytics logs were systematically parsed and mapped across multiple variables in R, culminating in a functional, filter-driven monitoring dashboard solution inside Tableau.

